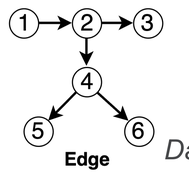
SCALABLE MULTI-NODE MULTI-GPU DATALOG ENGINE WITH ENERGY-AWARE PROFILING

Accelerating declarative analytics on HPC systems

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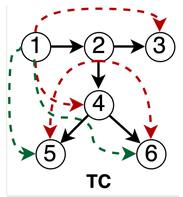
ABSTRACT

- **Datalog** simplifies large-scale analytics by expressing recursive logic with concise, declarative rules




```
TC(x, y) :- Edge(x, y).
TC(x, z) :- TC(x, y), Edge(y, z).
```

Datalog rules to compute Transitive Closure (TC) of a relation



- The rules are compiled down to **iterative relational algebra**

$$F_G(T) \triangleq G \cup \Pi_{1,2}(\rho_{0/1}(T) \bowtie_1 G)$$

- Datalog executes queries iteratively until a fixed point is reached, making it well-suited for **graph mining**, **deductive databases**, and **symbolic AI**
- **Parallelizing** these iterative relational operations on modern accelerators can greatly improve the performance of declarative analytics
- **MNMGDatalog** is the first *multi-node, multi-GPU Datalog engine* built on CUDA-Aware MPI
- Achieves up to **32× speedup** over distributed CPU-based engines on **Polaris**
- **Powerlog** enables GPU-level *energy profiling* for declarative workloads

MOTIVATION

Limitations of current Datalog Engines

Multi-threaded	Distributed (Apache Spark)	Multi-node Multi-threaded	Single-GPU	Multi-node Multi-GPUs (MNMG)
Soufflé	RDFox	SLOG	GPUJoin	?
LogicBlox	Radlog	PRAM	GPULog	
Scalability	Overhead	CPU-Only	VRAM	

Datalog on multi-node multi-GPU systems

High performance implementation for highly expressive language



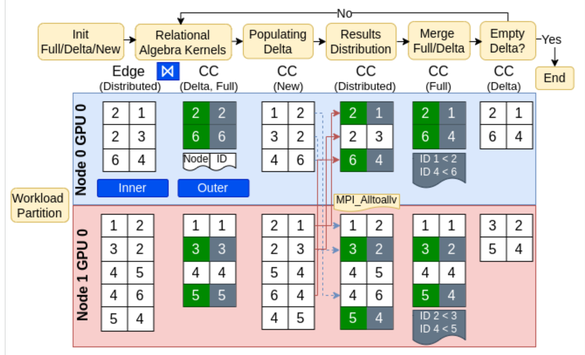
Requirements for MNMG Engine

1. Workload partitioning
2. Data representation
3. Efficient communication
4. Tuple materialization

METHODS

MNMGDatalog uses radix-hash partitioning and non-uniform all-to-all communication with GPU-aware hash tables for efficient tuple materialization

Materialization in fixed point iteration



Experiment environment

HW: AMD EPYC 7543P CPU, NVIDIA A100 GPUs with NVLink

Apps: Transitive Closure (TC), Same Generation (SG), Connected Component (CC)

Datasets: Stanford large network, SuiteSparse, Road network

MNMGDATALOG PERFORMANCE EVALUATION

We evaluate **MNMGDatalog** against state-of-the-art *single-GPU*, *shared-memory*, and *distributed multi-node* Datalog engines up to 32 NVIDIA A100 GPUs

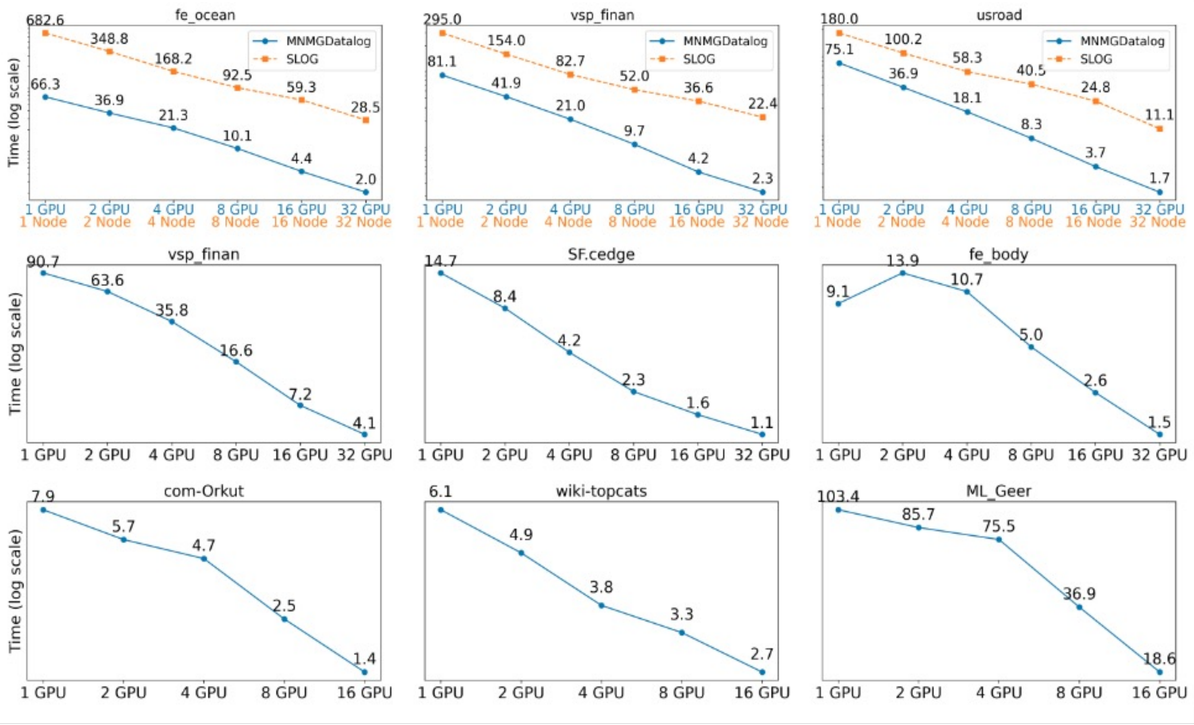
Single-GPU evaluation for TC

Dataset name	*TC* edges	Time (s) MNMGDatalog	GPULog	Soufflé	GPUJoin
com-dblp	1.91B	**13.58**	26.95	232.99	OOM
fe_ocean	1.67B	**66.34**	72.74	292.15	100.30
usroads	871M	**75.07**	78.08	222.76	364.55
vsp_finan	910M	**81.14**	82.75	239.33	125.94
Gnutella31	884M	**4.75**	7.64	96.82	OOM

Single-GPU evaluation for SG

Dataset name	*SG* size	Time (s) MNMGDatalog	GPULog	Soufflé	cuDF
fe_body	408M	**9.08**	18.41	74.26	OOM
loc-Brightkite	92.3M	**1.66**	11.67	48.18	OOM
fe_sphere	205M	**3.55**	7.88	48.12	OOM
CA-HepTH	74M	**0.60**	4.79	20.12	21.24

Multi-node evaluation for TC, SG, and CC



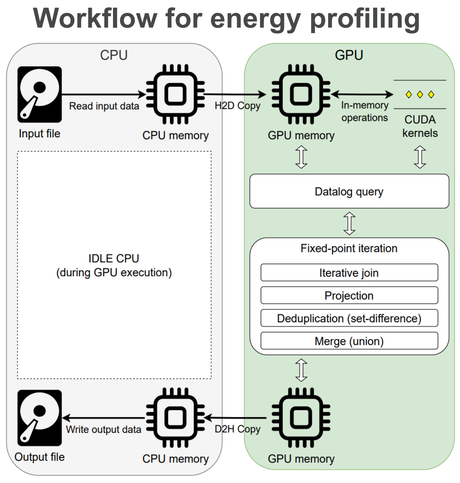
POWERLOG FOR ENERGY PROFILING

Evaluates energy-efficiency of SOTA GPU-powered engines on Polaris

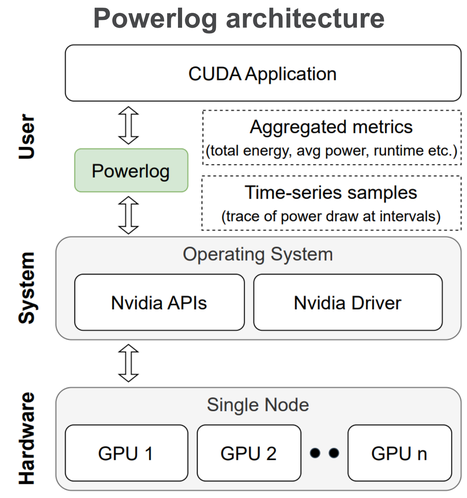
$$E = \int_0^T P(t)\,dt \;\approx\; \sum_{i=1}^{N} P_i \cdot \Delta t_i$$

where, E=total energy, N=# of samples, P_i=power draw at i, Δt_i=time between i, i+1

Workflow for energy profiling



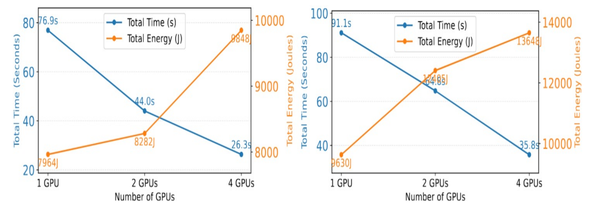
Powerlog architecture



Energy efficiency metric: Tuples/Joule

$$\frac{\text{Total number of tuples produced}}{\text{Total energy consumed (Joules)}}$$

MNMGDatalog multi-GPU power analysis



Comparison of GPU Datalog engines

Engine	Join Strategy	Data Structure
MNMGDATALOG	Radix hash join	GPU Hash table
INLJoin	Index nested loop join	GPU struct array
GPULOG	Sorted hash join	Hash-indexed sorted array
BJoin	Batch join with pooling	Hash-indexed sorted array
cuDF	Dataframe's merge	Python dataframe

Energy efficiency using Tuples/Joule

Application	Dataset	1 GPU	2 GPUs	4 GPUs
Transitive Closure	usroads	**109409**	105207	88483
Same Generation	vsp	**89802**	69710	63361

CONCLUSION

First multi-node multi-GPU Datalog engine

Single-GPU: Up to 7× speedup over GPULog
Multi-threaded: Up to 32× over Soufflé
Distributed: Up to 31.9× speedup over SLOG

- Introduces novel GPU-Aware communication for scalable recursive query evaluation
- Supports recursive aggregation for Datalog rules using high-throughput GPU kernels
- Power profiling of GPU-based Datalog engines

FUTURE PLAN

- Add spatial and temporal load balancing
- Develop GPU-Aware HIP and OneAPI implementations
- Benchmark on Neurosymbolic programming applications
- Extend **Powerlog** to multi-node environment
- Augment *Tuples/Joule* metric with Datalog-level counters (work/iteration, duplicate ratio, tuples materialized) and system-level metrics (HBM/DRAM bandwidth, cache hits/misses)

REFERENCES

- Gilray, T., Sahebolamri, A., Sun, Y., Kunapaneni, S., Kumar, S., & Micinski, K. (2024). Datalog with First-Class Facts. *VLDB*, *18*(3), 651-665.
- Jordan, H., Scholz, B., & Subotić, P. (2016, July). Soufflé: On synthesis of program analyzers. In *International Conference on Computer Aided Verification* (pp. 422-430).

MNMGDatalog evolution

Accelerating Datalog applications with cuDF — IA³ 2022 → Towards iterative relational algebra on the {GPU} — USENIX ATC '23 → Optimizing Datalog for the GPU — ASPLOS 2025 → Multi-node multi-GPU Datalog — ICS

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